ROBIN BILODEAU

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EDUCATION

University of Ottawa — **Ottawa, Ontario, Canada**
Bachelor of Science in Computer Science and Mathematics — Sep 2024 – Apr 2028

- **Relevant Courses**: Intro to Computing 1, Intro to Computing 2, Linear Algebra, Differential Equations and Laplace Transforms, Discrete Structures, Elementary Real Analysis, Introduction to probability

WORK EXPERIENCE

Scotiabank - Global Markets — **Toronto, Ontario, Canada**
Quantitative Analyst Intern — Sep 2025 – Dec 2025

- Refactored a legacy pricing process into a modular architecture supporting multiple rates derivatives, including swaps, swaptions, caps, and floors
- Optimized performance bottlenecks in core analytical functions, leading to substantial runtime improvements
- Conducted in-depth testing and investigation of pricing models to identify product-dependent discrepancies and assess model validity, and visualized discrepancy differences across products to communicate findings clearly

RESEARCH EXPERIENCE

Privacy-Preserving Zero-Knowledge Credential Protocol — **Ottawa, Ontario, Canada**
Undergraduate Researcher — Feb 2026 – Present

- Designed and implemented a zero-knowledge proof system over Pedersen commitments enabling privacy-preserving equality testing on hidden attributes
- Migrated the protocol's key exchange and signatures to post-quantum primitives (Kyber768 KEM, Dilithium3), replacing classical Diffie-Hellman and Ed25519 while preserving mutual authentication

A Data-Driven Assessment of Machine Learning Models for Climate-Sensitive Wildfire Susceptibility in Alberta — **Ottawa, Ontario, Canada**
Undergraduate Researcher — Feb 2026 – Present

- Developing and extending the GMDH algorithm to produce an interpretable, self-organizing machine learning pipeline for wildfire susceptibility detection
- Abstract accepted to Earth 2026 Conference

Uncertainty-Aware Statistical Extensions for HDBSCAN Clustering — **Ottawa, Ontario, Canada**
Undergraduate Researcher — Apr 2025 – Oct 2025

- Developed a Monte Carlo sampling framework to repeatedly adjust probability rank matrices, yielding statistically stable core-distance estimates for clustering high-noise data leading to reduced sensitivity to outliers and improved cluster detection reliability
- Coordinated a fast Fourier-based deconvolution estimator to recover underlying data distributions from noise-contaminated observations, enhancing core distance estimates for robust HDBSCAN clustering

LEADERSHIP AND ACTIVITIES

SSC Case Study Competition - Parkinson's Disease Classification — **Hamilton, Ontario, Canada**
Team Member — Apr 2026 – Jun 2026

- Performed end-to-end data cleaning on a high-dimensional clinical dataset (500+ variables) from the Canadian Open Parkinson Network (C-OPN), including leakage analysis to identify and remove variables that directly encode diagnostic criteria for PD vs. atypical parkinsonism, and outcome-stratified MICE imputation to handle missing data
- Developed a random forest classifier achieving greater than 95% accuracy in distinguishing Parkinson's disease from atypical parkinsonism syndromes, using stratified k-fold cross-validation with fold splitting prior to imputation

SKILLS, LANGUAGES, INTERESTS

- **Languages**: English (Native speaker), French (Native speaker)
- **Programming**: Python, R, Java, Typst, C++ (Beginner)
- **Tools**: Microsoft Word, Git, Visual Studio Code, Canva, Jupyter
- **Libraries**: matplotlib, numpy, Scikit-learn, pandas, yfinance
- **Athletics**: Boxing, Hockey, Soccer, Volleyball
- **Activities**: Tutoring, Mathematics Club, Undergraduate Research Club